

Quick Check

Lesson 4: Liver and Skin: Hidden Waste Processors Behind the Hot Afternoon

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.3: Excretion and Homeostasis in Animals | Lesson 4

Under the microscope, liver tissue shows wide, thin-walled sinusoids running between rows of hepatocytes (liver cells). What does this structure allow?

- A** Sweat to be made in the liver and carried to the skin
- B** Blood to pass the hepatocytes, so wastes and other substances can move easily between blood and liver cells
- C** Blood to bypass the hepatocytes so that they are protected from wastes
- D** Urine to collect between the liver cells before it leaves the body

A section of human skin shows a coiled sweat gland surrounded by many blood capillaries. Why are the capillaries important to the sweat gland?

- A** They make urea from amino acids inside the skin
- B** They store the sweat until the body needs it
- C** They carry sweat directly out of the pore onto the skin surface
- D** They bring blood to the gland, which supplies the water, salts and a little urea that make up sweat

A goat eats dry grass that is rich in protein. Some amino acids are in excess of what the body needs. What happens to these excess amino acids in the liver?

- A** They are changed directly into bile and sent to the intestine
- B** They lose their amino group in deamination, which forms ammonia and then urea
- C** They are filtered out unchanged by the liver and stored as urine
- D** They are changed into sweat and lost through the skin

Old red blood cells are broken down in the body. The pigment bilirubin from the haemoglobin is removed by the liver. Where does this bilirubin go?

- A** It goes into the blood and is removed by the kidney as urea
- B** It is lost through the sweat glands in sweat
- C** It is excreted in bile, which passes into the intestine and leaves with the faeces
- D** It is stored permanently in the hepatocytes

In cirrhosis, scar tissue damages hepatocytes and reduces blood flow through the liver. Which result is expected?

- A** The kidneys start to make bilirubin to replace the liver
- B** Less urea and less bilirubin in the blood, because the liver makes more bile
- C** Urea and bilirubin build up in the blood, and the skin and eyes may turn yellow (jaundice)
- D** The skin makes more sweat, which removes all the wastes

On a hot afternoon, a human and a horse lose heat by sweating, while a dog pants. Which statement best explains why the dog pants?

- A** A dog has no blood capillaries near the skin
- B** A dog pants to make more bile for digestion
- C** A dog has few sweat glands, so it loses heat by evaporation from its tongue and mouth instead
- D** A dog makes sweat that is only urea, which cannot cool the body

A goat's urine contains urea. Which sequence correctly shows where the urea comes from and how it leaves the body?

- A** Amino acids go to the liver, where urea is made, and then the blood carries it to the kidney, which removes it in urine
- B** Amino acids go to the kidney, where urea is made, and then to the liver, which removes it in urine
- C** Amino acids go to the skin, where urea is made, and then to the kidney
- D** Haemoglobin goes to the liver, where urea is made, and then to the intestine as bile

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).